```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re

transaction = pd.read_excel("QVI_transaction_data.xlsx")
customer = pd.read_csv("QVI_purchase_behaviour.csv")

transaction.head()
customer.head()
```

```
   LYLTY_CARD_NBR               LIFESTAGE PREMIUM_CUSTOMER
0            1000   YOUNG SINGLES/COUPLES          Premium
1            1002   YOUNG SINGLES/COUPLES       Mainstream
2            1003          YOUNG FAMILIES           Budget
3            1004   OLDER SINGLES/COUPLES       Mainstream
4            1005  MIDAGE SINGLES/COUPLES       Mainstream
```

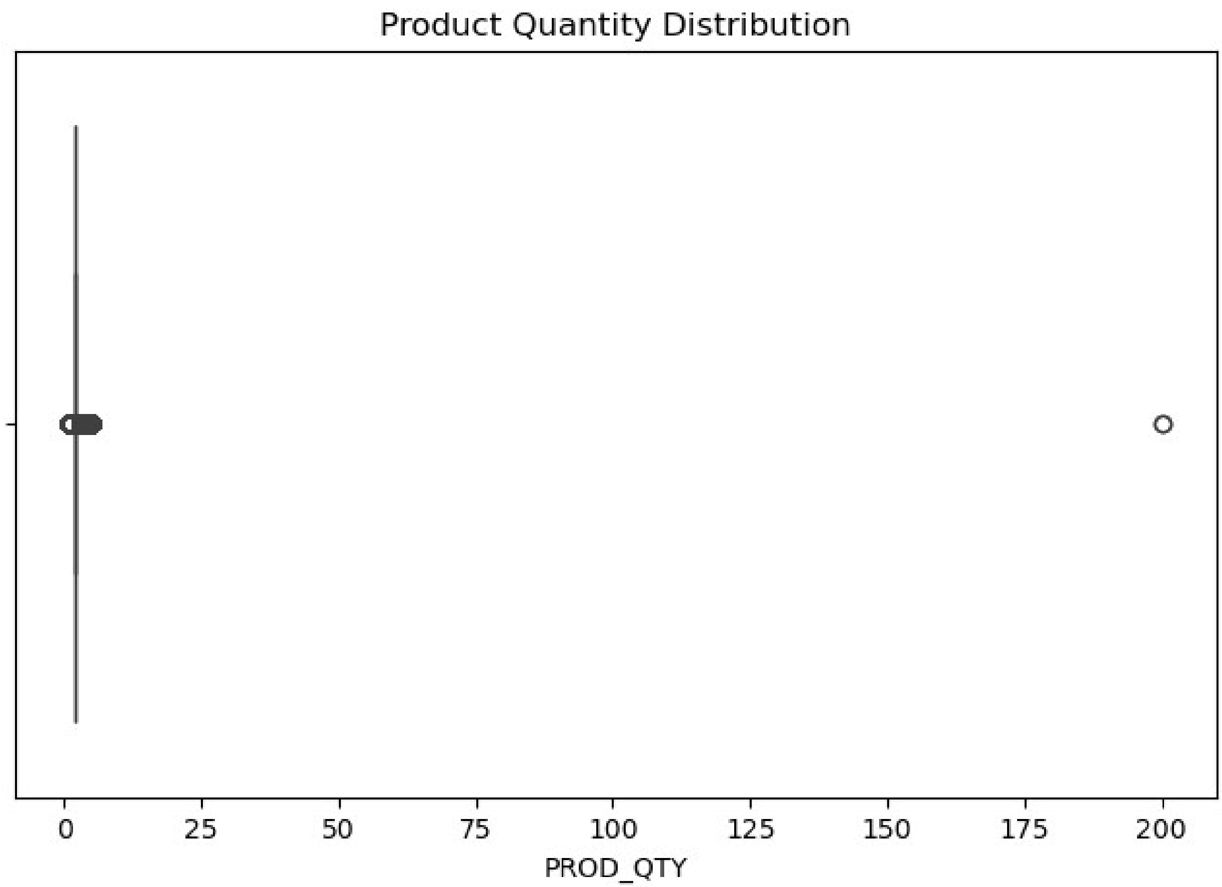
```python
transaction.info()
customer.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   DATE            264836 non-null  int64
 1   STORE_NBR       264836 non-null  int64
 2   LYLTY_CARD_NBR  264836 non-null  int64
 3   TXN_ID          264836 non-null  int64
 4   PROD_NBR        264836 non-null  int64
 5   PROD_NAME       264836 non-null  object
 6   PROD_QTY        264836 non-null  int64
 7   TOT_SALES       264836 non-null  float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   LYLTY_CARD_NBR    72637 non-null  int64
 1   LIFESTAGE         72637 non-null  object
 2   PREMIUM_CUSTOMER  72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

```python
transaction.isnull().sum()
customer.isnull().sum()
```

```
LYLTY_CARD_NBR      0
LIFESTAGE           0
PREMIUM_CUSTOMER    0
dtype: int64

transaction.duplicated().sum()
customer.duplicated().sum()

np.int64(0)

data = pd.merge(
    left=transaction,
    right=customer,
    on="LYLTY_CARD_NBR",
    how="left"
)

data.head()

        DATE  STORE_NBR  LYLTY_CARD_NBR  TXN_ID  PROD_NBR  \
0 2018-10-17          1            1000       1         5
1 2019-05-14          1            1307     348        66
2 2019-05-20          1            1343     383        61
3 2018-08-17          2            2373     974        69
4 2018-08-18          2            2426    1038       108

                                  PROD_NAME  PROD_QTY  TOT_SALES  \
0    Natural Chip        Compny SeaSalt175g         2        6.0
1                  CCs Nacho Cheese    175g         3        6.3
2    Smiths Crinkle Cut  Chips Chicken 170g         2        2.9
3    Smiths Chip Thinly  S/Cream&Onion 175g         5       15.0
4  Kettle Tortilla ChpsHny&Jlpno Chili 150g         3       13.8

                LIFESTAGE PREMIUM_CUSTOMER
0   YOUNG SINGLES/COUPLES          Premium
1  MIDAGE SINGLES/COUPLES           Budget
2  MIDAGE SINGLES/COUPLES           Budget
3  MIDAGE SINGLES/COUPLES           Budget
4  MIDAGE SINGLES/COUPLES           Budget

print(transaction["PROD_QTY"].describe())

plt.figure(figsize=(8,5))
sns.boxplot(x=transaction["PROD_QTY"])
plt.title("Product Quantity Distribution")
plt.show()

transaction = transaction[transaction["PROD_QTY"] < 6]

print(transaction["PROD_QTY"].describe())
```

```
count    264836.000000
mean          1.907309
std           0.643654
min           1.000000
25%           2.000000
50%           2.000000
75%           2.000000
max         200.000000
Name: PROD_QTY, dtype: float64
```




```
count    264834.000000
mean          1.905813
std           0.343436
min           1.000000
25%           2.000000
50%           2.000000
75%           2.000000
max           5.000000
Name: PROD_QTY, dtype: float64
```

```
transaction["PACK_SIZE"] = (
    transaction["PROD_NAME"]
```

```python
    .str.extract(r'(\d+)')
    .astype(int)
)

transaction["BRAND"] = (
    transaction["PROD_NAME"]
    .str.split()
    .str[0]
)

transaction["BRAND"] = transaction["BRAND"].replace({
    "Dorito":"Doritos",
    "Grain":"GrainWaves",
    "GrnWves":"GrainWaves",
    "Infzns":"Infuzions",
    "Red":"RRD",
    "Smith":"Smiths",
    "Snbts":"Sunbites",
    "WW":"Woolworths"
})

print(transaction[["PROD_NAME","BRAND","PACK_SIZE"]].head())
```

```
                                  PROD_NAME    BRAND  PACK_SIZE
0    Natural Chip        Compny SeaSalt175g  Natural        175
1                  CCs Nacho Cheese    175g      CCs        175
2    Smiths Crinkle Cut  Chips Chicken 170g   Smiths        170
3    Smiths Chip Thinly  S/Cream&Onion 175g   Smiths        175
4  Kettle Tortilla ChpsHny&Jlpno Chili 150g   Kettle        150
```

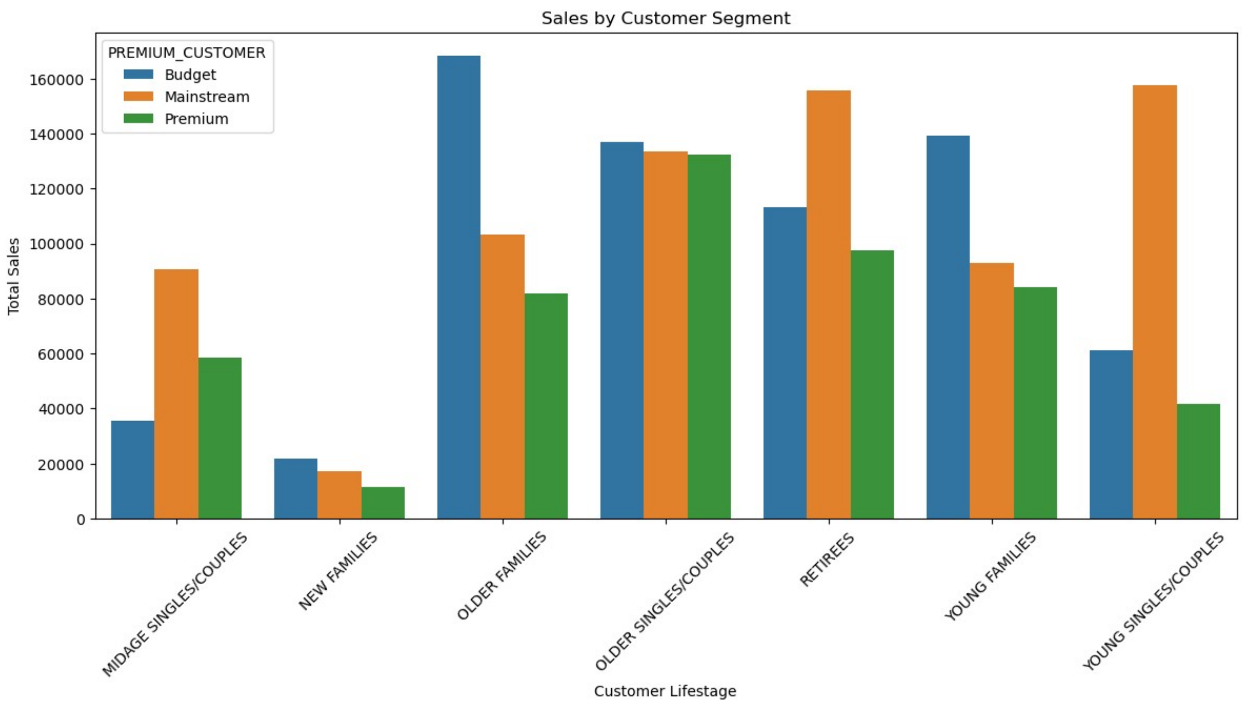
```python
segment = data.groupby(
    ["LIFESTAGE","PREMIUM_CUSTOMER"]
).agg(

    Total_Sales=("TOT_SALES","sum"),
    Customers=("LYLTY_CARD_NBR","nunique"),
    Transactions=("TXN_ID","count"),
    Quantity=("PROD_QTY","sum")

).reset_index()

segment["Avg Spend per Customer"] = (
    segment["Total_Sales"] /
    segment["Customers"]
)

segment["Avg Units per Customer"] = (
    segment["Quantity"] /
    segment["Customers"]
)
```

```python
segment["Avg Price per Unit"] = (
    segment["Total_Sales"] /
    segment["Quantity"]
)

segment
```

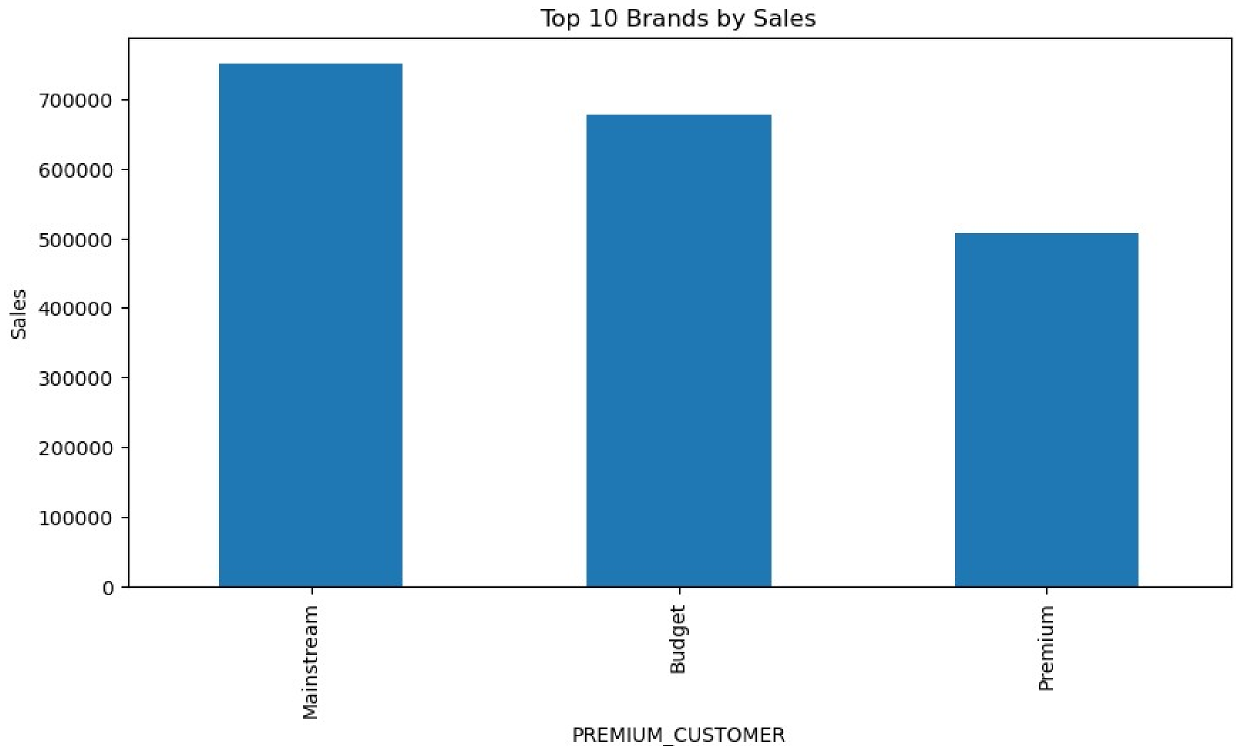
```
                 LIFESTAGE PREMIUM_CUSTOMER  Total_Sales  Customers  \
0   MIDAGE SINGLES/COUPLES           Budget     35514.80       1504
1   MIDAGE SINGLES/COUPLES       Mainstream     90803.85       3340
2   MIDAGE SINGLES/COUPLES          Premium     58432.65       2431
3             NEW FAMILIES           Budget     21928.45       1112
4             NEW FAMILIES       Mainstream     17013.90        849
5             NEW FAMILIES          Premium     11491.10        588
6           OLDER FAMILIES           Budget    168363.25       4675
7           OLDER FAMILIES       Mainstream    103445.55       2831
8           OLDER FAMILIES          Premium     81958.40       2274
9    OLDER SINGLES/COUPLES           Budget    136769.80       4929
10   OLDER SINGLES/COUPLES       Mainstream    133393.80       4930
11   OLDER SINGLES/COUPLES          Premium    132263.15       4750
12                RETIREES           Budget    113147.80       4454
13                RETIREES       Mainstream    155677.05       6479
14                RETIREES          Premium     97646.05       3872
15          YOUNG FAMILIES           Budget    139345.85       4017
16          YOUNG FAMILIES       Mainstream     92788.75       2728
17          YOUNG FAMILIES          Premium     84025.50       2433
18   YOUNG SINGLES/COUPLES           Budget     61141.60       3779
19   YOUNG SINGLES/COUPLES       Mainstream    157621.60       8088
20   YOUNG SINGLES/COUPLES          Premium     41642.10       2574

    Transactions  Quantity  Avg Spend per Customer  Avg Units per
Customer  \
0           5020      9496               23.613564
6.313830
1          11874     22699               27.186781
6.796108
2           8216     15526               24.036466
6.386672
3           3005      5571               19.719829
5.009892
4           2325      4319               20.039929
5.087161
5           1589      2957               19.542687
5.028912
6          23160     45065               36.013529
9.639572
7          14244     27756               36.540286
9.804309
8          11192     22171               36.041513
```

```
9.749780
9           18407     35220                     27.747981
7.145466
10          18318     34997                     27.057566
7.098783
11          17754     33986                     27.844874
7.154947
12          15201     28764                     25.403637
6.458015
13          21466     40518                     24.027944
6.253743
14          13096     24884                     25.218505
6.426653
15          19122     37111                     34.689034
9.238486
16          12907     25044                     34.013471
9.180352
17          11563     22406                     34.535758
9.209207
18           9242     16671                     16.179307
4.411485
19          20854     38632                     19.488328
4.776459
20           6281     11331                     16.177972
4.402098

    Avg Price per Unit
0             3.739975
1             4.000346
2             3.763535
3             3.936178
4             3.939315
5             3.886067
6             3.736009
7             3.726962
8             3.696649
9             3.883299
10            3.811578
11            3.891695
12            3.933660
13            3.842170
14            3.924050
15            3.754840
16            3.705029
17            3.750134
18            3.667542
19            4.080079
20            3.675060
```

```
plt.figure(figsize=(14,6))

sns.barplot(
    data=segment,
    x="LIFESTAGE",
    y="Total_Sales",
    hue="PREMIUM_CUSTOMER"
)

plt.xticks(rotation=45)
plt.title("Sales by Customer Segment")
plt.xlabel("Customer Lifestage")
plt.ylabel("Total Sales")
plt.show()
```




```
brand_sales = (
    data.groupby("PREMIUM_CUSTOMER")["TOT_SALES"]
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10,5))

brand_sales.head(10).plot(kind="bar")

plt.title("Top Premium custoemr by Sales")
plt.ylabel("Sales")
plt.show()
```




```
Business Insights

1. Mainstream Young Singles/Couples contribute the highest total chip
sales.

2. Premium customers spend more on average compared to Budget
customers.

3. Larger pack sizes generate higher revenue than smaller packs.

4. Kettle, Smiths, Doritos, and Pringles are the best-performing
brands.

5. Customer lifestage has a significant impact on purchasing
behaviour.

Recommendations

• Increase promotions targeting Mainstream Young Singles/Couples.
• Focus shelf space on high-performing brands.
• Promote larger pack sizes through bundle offers.
• Introduce loyalty rewards for Premium customers.
• Use personalized marketing campaigns based on customer lifestage.
```